Quiz 5

Continuous laws and joint support

Try each question before reading the worked solutions on page 2.

1. $f_X(x) = cx$ on $0 < x < 2$, zero elsewhere. What is c?

A. $1/2$

B. 1

C. 2

D. $1/4$

2. A variable has a density with $f_X(1) = 2$. What is $P(X = 1)$?

A. 2

B. $1/2$

C. 0

D. Cannot be determined.

3. $T \sim \text{Exp}(3)$ with time in minutes. What is $P(T > 1)$?

A. $1 - e^{-3}$

B. e^{-3}

C. $e^{-1/3}$

D. $3e^{-3}$

4. $X \sim N(20, 9)$. Which is $P(X > 23)$?

A. $\Phi(1)$

B. $1 - \Phi(1/3)$

C. $1 - \Phi(3)$

D. $1 - \Phi(1)$

5. The joint density is 2 on $0 < x < y < 1$. What is $f_X(x)$ for $0 < x < 1$?

A. $2x$

B. 2

C. $2(1 - x)$

D. $1 - x$

Worked solutions

1. Normalizing a density

Answer A

A density must integrate to one over its support. Using u as the integration variable,

$$1 = \int_0^2 cu\,du = c\left[\frac{u^2}{2}\right]_0^2 = 2c \quad \Longrightarrow \quad c = \frac{1}{2}.$$

Thus $f_X(x) = x/2$ for $0 < x < 2$, and $f_X(x) = 0$ elsewhere. This density is nonnegative.

2. Density height and probability

Answer C

For a variable with a density, probability is area under the density. A single point has zero width, so

$$P(X = 1) = \int_1^1 f_X(u)\,du = 0.$$

The value $f_X(1) = 2$ is a density height, not a point probability. Density heights may exceed one.

3. Exponential right tail

Answer B

The parameter 3 is the rate per minute. Hence $f_T(t) = 3e^{-3t}$ for $t > 0$, and zero elsewhere. Integrate over the requested right tail:

$$P(T > 1) = \int_1^\infty 3e^{-3u}\,du = \left[-e^{-3u}\right]_1^\infty = e^{-3}.$$

Equivalently, use $P(T > t) = e^{-3t}$ for $t \geq 0$.

4. Standardizing a normal variable

Answer D

In $N(\mu, \sigma^2)$, the second parameter is the variance. Here $\sigma = \sqrt{9} = 3$, so $Z = (X - 20)/3 \sim N(0, 1)$. Therefore

$$P(X > 23) = P\left(Z > \frac{23 - 20}{3}\right) = P(Z > 1) = 1 - \Phi(1).$$

The standard normal CDF is $\Phi(z) = P(Z \leq z)$; subtract it from one for a right tail.

5. Support determines the integral

Answer C

Fix $x \in (0, 1)$. The joint support allows $x < y < 1$, so integrate out Y using the dummy variable v:

$$f_X(x) = \int_x^1 f_{X,Y}(x, v)\,dv = \int_x^1 2\,dv = 2(1 - x).$$

For $x \notin (0, 1)$, $f_X(x) = 0$. Integrating from 0 to 1 at every x would include points outside the joint support.